

Day 5 – List programs

1. Create a List and Perform CRUD Operations
2. Find Maximum and Minimum in a List (Without `max()` and `min()`)
3. Remove Duplicates from a List
4. Find Second Largest Element
5. Remove Duplicates (Using Nested Loops)
6. Left Rotate List by One Position
7. Right Rotate List by One Position
8. Count Occurrences of an Element
9. Search an Element in a List

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1. Create list and perform CRUD operation

```
numbers = [10, 20, 30, 40]
print("Original List:", numbers)
numbers.append(50)
print("After Add:", numbers)
print("First Element:", numbers[0])
numbers[1] = 25
print("After Update:", numbers)
numbers.remove(30)
print("After Delete:", numbers)
```

2. Find Maximum and Minimum in a List (Without max() and min())

```
numbers = [12, 45, 7, 89, 23]
largest = numbers[0]
smallest = numbers[0]
for i in numbers:
    if i > largest:
        largest = i
    if i < smallest:
        smallest = i
print("Maximum =", largest)
print("Minimum =", smallest)
```

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3.Remove Duplicate from a list

```
numbers = [1, 2, 3, 2, 4, 5, 3, 1]
```

```
result = []
```

```
for i in numbers:
```

```
    if i not in result:
```

```
        result.append(i)
```

```
print(result)
```

4.Find second largest element in a list

```
numbers = [12, 45, 7, 89, 34]
```

```
largest = numbers[0]
```

```
second = numbers[0]
```

```
for i in numbers:
```

```
    if i > largest:
```

```
        second = largest
```

```
        largest = i
```

```
    elif i > second and i != largest:
```

```
        second = i
```

```
print("Second Largest =", second)
```

5.Remove duplicate

```
numbers = [1, 2, 2, 3, 4, 3, 5]
```

```
result = []
```

```
for i in numbers:
```

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```
found = False
for j in result:
    if i == j:
        found = True
        break
if not found:
    result.append(i)
print(result)
```

6.Left rotate by one position

```
numbers = [10, 20, 30, 40, 50]
first = numbers[0]
for i in range(len(numbers) - 1):
    numbers[i] = numbers[i + 1]
numbers[-1] = first
print(numbers)
```

7.Right rotate by one position

```
numbers = [10, 20, 30, 40, 50]
last = numbers[-1]
for i in range(len(numbers) - 1, 0, -1):
    numbers[i] = numbers[i - 1]
numbers[0] = last
print(numbers)
```

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8.Count occurrence of an element

```
numbers = [1, 2, 3, 2, 4, 2, 5]
```

```
key = 2
```

```
count = 0
```

```
for i in numbers:
```

```
    if i == key:
```

```
        count += 1
```

```
print("Count =", count)
```

9.Search an element in a list

```
numbers = [10, 20, 30, 40, 50]
```

```
key = int(input("Enter element to search: "))
```

```
found = False
```

```
for i in numbers:
```

```
    if i == key:
```

```
        found = True
```

```
        break
```

```
if found:
```

```
    print("Element Found")
```

```
else:
```

```
    print("Element Not Found")
```